



DO-IT-YOURSELF: PROJECT

read from and write to your channel.

Data Import/Export. This tab enables you to import and export channel data.

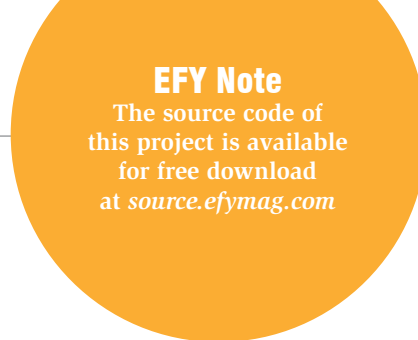
Fig. 6: Entering channel setting values

You can thus create a private channel as per your requirement and then use the provided API to write the data that is to be exported in different formats. The data can be extracted in all three formats: xml, csv, and json. This data can be used in Android based apps, as json format is available, and also for web applications. The requisites like SSID code, password, channel name, and the API key should be changed while you create a new system for a different person.

Coding

After the ThingSpeak API and channel setting is done, create the code to read sensors' data and upload it to ThingSpeak cloud network in real time. In the code, you need to do the SSID and password settings for your network, which can be your phone hotspot or home Wi-Fi network.

For uploading the source



EFY Note

The source code of this project is available for free download at source.efymag.com

code *smart_ring.ino* into Arduino Uno board, connect the Arduino board to desktop/laptop using USB cable. Next, select the board and port and upload the code (*smart_ring.ino*). You can download it from the link <https://www.electronicsforu.com/electronics-projects/smart-wearable-ring>

Construction and testing

Provide power to the assembled circuit with a 5V USB adaptor or any other 5V DC regulated power supply and then wait for a few seconds, till the circuit connects to Wi-Fi. Now open the ThingSpeak channel you have created; you can see your health data in real time.

When you place a finger on the MAX30100 sensor, it will start showing your pulse rate, oxygen percentage in blood, and temperature of your body in real time on the ThingSpeak channel. When you press the SOS button, you can see the state of SOS button in real time from any device anywhere.

This futuristic device can be useful for everyone, especially so for the old and infirm, the sick and bed-ridden, and the pregnant women, all of whom need special care and attention. It can reduce their trips to the doctor or a hospital as the vital health parameters can now be viewed remotely. Welcome to the new age of health monitoring!

During assembly, the ESP8266 NodeMCU is replaced with ESP 8266 chip and all other sensors and components can be fixed into the small ring, as shown in the Fig. 8. **EFY**

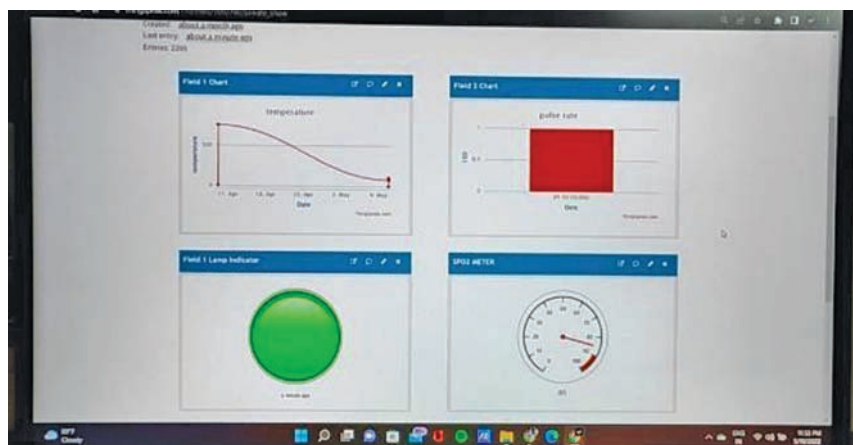


Fig. 7: The health data as seen on a laptop

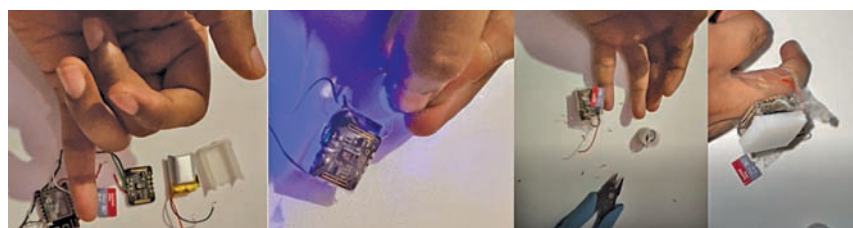


Fig. 8: Prototype of ring assembly

Duraiarasu.E is B.E in Electronics and Communications Engineering (ECE). His areas of interest include Embedded Systems, VLSI, IoT, Robotics and Automation, and Machine Learning